



Dev Board

A single-board computer with a removable system-on-module (SoM) featuring the Edge TPU.

- **Supported OS:** Mendel Linux (derivative of Debian)
- **Supported Framework:** TensorFlow Lite, AutoML Edge
- **Languages:** Python and C++
- **CPU:** NXP i.MX 8M SoC (quad Cortex-A53, Cortex-M4F)
- **GPU:** Integrated GC7000 Lite Graphics
- **ML accelerator:** Google Edge TPU coprocessor
- **RAM:** 1 GB LPDDR4
- **Flash memory:** 8 GB eMMC
- **Wireless:** Wi-Fi 2x2 MIMO (802.11b/g/n/ac 2.4/5GHz) and Bluetooth 4.2
- **Dimensions:** 48mm x 40mm x 5mm



USB Accelerator

A USB accessory featuring the Edge TPU that brings ML inferencing to existing systems.

- **Supported host OS:** Debian Linux
- **Compatible with:** Raspberry Pi boards
- **Supported Framework:** TensorFlow Lite, AutoML Vision Edge.
- **ML accelerator:** Google Edge TPU coprocessor
- **Connector:** USB 3.0 Type-C* (data/power)
- **Dimensions:** 65 mm x 30 mm

Figure 2: Coral products with edge TPU (Image source: <http://coral.withgoogle.com>)

layers may get faster by 10 to 50 per cent on the CPU. Fully connected and RNN layers may speed up operation by 3x.

TF Lite now supports post-training quantisation, which reduces the size along with compute demands greatly. TensorFlow 2.0 offers simplified APIs to build models with quantisation and by pruning optimisations.

A normal dense layer without quantisation looks like what follows:

```
tf.keras.layers.Dense(512, activation='relu')
```

Whereas a quantised dense layer looks like what's shown below:

```
quantize.Quantize(tf.keras.layers.Dense(512,
activation='relu'))
```

Pruning is a technique used to drop connections that are ineffective. In general, 'dense' layers contain lots of connections which don't influence the output. Such connections can be dropped by making the weight zero. Tensors with lots of zeros may be represented as 'sparse' and can be compressed. Also, the number of operations in a sparse tensor is less.

Building a layer with *prune* is as simple as using the following command:

```
prune.Prune(tf.keras.layers.Dense(512, activation='relu'))
```

In a pipeline, there is Keras based quantised training and

Keras based connection pruning. These optimisations may push TF Lite further ahead of the competition, with regard to other frameworks.

Coral

Coral is a new platform for creating products with on-device ML acceleration. The first product here features Google's Edge TPU in SBC and USB form factors. TensorFlow Lite is officially supported on this platform, with the salient features being very fast inference speed, privacy and no reliance on network connection.

More details related to hardware specifications, pricing, and a getting started guide can be found at <https://coral.withgoogle.com>.

With these advances as well as a wider ecosystem, it's very evident that TensorFlow may become the leading framework for artificial intelligence and machine learning, similar to how Android evolved in the mobile world. **END** 🐧

References

- [1] <https://www.tensorflow.org/dev-summit>
- [2] <http://www.tensorflow.org>
- [3] <https://coral.withgoogle.com>

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The author is keenly interested in researching open source technologies like Android, artificial intelligence and deep learning.